MA1522 Linear Algebra for Computing
L2 Lesson 5



Tan Kai Meng

Semester 1, 2026/27

LU Factorization

Let $\boldsymbol{A}$ be a matrix. To obtain a row echelon form $\boldsymbol{R}$ of $\boldsymbol{A}$, we can perform Gaussian elimination to $\boldsymbol{A}$.

Recall that Gaussian elimination involves only elementary row operations of the form $R_i \leftrightarrow R_j$ and $R_i + cR_j$ for $i > j$.

Assume that $\boldsymbol{A}$ requires only those operations of the form $R_i + cR_j$ for $i > j$:

$$\boldsymbol{A} \xrightarrow[\boldsymbol{E}_1]{R_{i_1}+c_1R_{j_1}} \bullet \xrightarrow[\boldsymbol{E}_2]{R_{i_2}+c_2R_{j_2}} \bullet \cdots \bullet \xrightarrow[\boldsymbol{E}_r]{R_{i_r}+c_rR_{j_r}} \boldsymbol{R},$$

where $i_1 > j_1$, $i_2 > j_2$, $\ldots$, $i_r > j_r$. Then $\boldsymbol{R} = \boldsymbol{E}_r \cdots \boldsymbol{E}_2\boldsymbol{E}_1\boldsymbol{A}$, and hence

$$\boldsymbol{A} = \boldsymbol{E}_1^{-1}\boldsymbol{E}_2^{-1} \cdots \boldsymbol{E}_r^{-1}\boldsymbol{R} = \boldsymbol{L}\boldsymbol{R},$$

where $\boldsymbol{L} = \boldsymbol{E}_1^{-1}\boldsymbol{E}_2^{-1} \cdots \boldsymbol{E}_r^{-1}$.

Note that each $\boldsymbol{E}_a^{-1}$ is the elementary matrix corresponding to $R_{i_a} - c_aR_{j_a}$ where $i_a > j_a$, so that it is **unit lower triangular**. Thus L is **unit lower triangular**.

$\boldsymbol{A} = \boldsymbol{L}\boldsymbol{R}$ is called the **LU factorization of $\boldsymbol{A}$**.

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$A = LR$ is called the **LU factorization of A**.

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Question 1

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Compute the LU factorization of $\begin{pmatrix} 2 & -1 & 2 \\ -6 & 0 & -2 \\ 8 & -1 & 5 \end{pmatrix}$.

(See Pg 125–127 of Chapter 2 Lecture Slides.)

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Solution.

$$\begin{pmatrix} 2 & -1 & 2 \\ -6 & 0 & -2 \\ 8 & -1 & 5 \end{pmatrix} \xrightarrow[R_3-4R_1]{R_2+3R_1} \begin{pmatrix} 2 & -1 & 2 \\ 0 & -3 & 4 \\ 0 & 3 & -3 \end{pmatrix} \xrightarrow{R_3+R_2} \begin{pmatrix} 2 & -1 & 2 \\ 0 & -3 & 4 \\ 0 & 0 & 1 \end{pmatrix}.$$

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□

Using LU Factorization to Solve Matrix Equations

If $\boldsymbol{A} = \boldsymbol{LR}$ is an LU factorization of $\boldsymbol{A}$, to solve $\boldsymbol{Ax} = \boldsymbol{b}$, let $\boldsymbol{y} = \boldsymbol{Rx}$ so that

$$\boldsymbol{Ly} = \boldsymbol{L}(\boldsymbol{Rx}) = (\boldsymbol{LR})\boldsymbol{x} = \boldsymbol{Ax} = \boldsymbol{b},$$

and solve

$$\begin{aligned} \boldsymbol{Ly} &= \boldsymbol{b}; \\ \boldsymbol{Rx} &= \boldsymbol{y}. \end{aligned}$$

As L and R are 'triangular', the latter matrix equations can be very easily solved.

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As $\boldsymbol{L}$ and $\boldsymbol{R}$ are 'triangular', the latter matrix equations can be very easily solved.

Question 2

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Find the LU factorization for the matrix $\boldsymbol{A}$, and solve the equation $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$:

a. $\boldsymbol{A} = \begin{pmatrix} 2 & -4 & 4 & -2 \\ 6 & -9 & 7 & -3 \\ -1 & -4 & 8 & 0 \end{pmatrix}, \quad \boldsymbol{b} = \begin{pmatrix} 0 \\ 0 \\ 5 \end{pmatrix}.$

b. $\boldsymbol{A} = \begin{pmatrix} 2 & -6 & 6 \\ -4 & 5 & -7 \\ 3 & 5 & -1 \\ -6 & 4 & -8 \\ 8 & -3 & 9 \end{pmatrix}, \quad \boldsymbol{b} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}.$

(Cf. Pg 121 of Chapter 2 Lecture Slides.)

Question 2(a): Solution

$$\boldsymbol{A} = \begin{pmatrix} 2 & -4 & 4 & -2 \\ 6 & -9 & 7 & -3 \\ -1 & -4 & 8 & 0 \end{pmatrix} \xrightarrow[R_3+\frac{1}{2}R_1]{R_2-3R_1} \begin{pmatrix} 2 & -4 & 4 & -2 \\ 0 & 3 & -5 & 3 \\ 0 & -6 & 10 & -1 \end{pmatrix} \xrightarrow{R_3+2R_2} \begin{pmatrix} 2 & -4 & 4 & -2 \\ 0 & 3 & -5 & 3 \\ 0 & 0 & 0 & 5 \end{pmatrix} = \boldsymbol{R}.$$

Thus $\begin{pmatrix} 2 & -4 & 4 & -2 \\ 6 & -9 & 7 & -3 \\ -1 & -4 & 8 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ -\frac{1}{2} & -2 & 1 \end{pmatrix} \begin{pmatrix} 2 & -4 & 4 & -2 \\ 0 & 3 & -5 & 3 \\ 0 & 0 & 0 & 5 \end{pmatrix}$ is the LU factorization of $\boldsymbol{A}$.

Let $\boldsymbol{y} = \boldsymbol{R}\boldsymbol{x}$. Then

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \Leftrightarrow \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ -\frac{1}{2} & -2 & 1 \end{pmatrix} \boldsymbol{y} = \begin{pmatrix} 0 \\ 0 \\ 5 \end{pmatrix} \Leftrightarrow \boldsymbol{y} = \begin{pmatrix} 0 \\ 0 \\ 5 \end{pmatrix};$$

$$\boldsymbol{R}\boldsymbol{x} = \boldsymbol{y} \Leftrightarrow \begin{pmatrix} 2 & -4 & 4 & -2 \\ 0 & 3 & -5 & 3 \\ 0 & 0 & 0 & 5 \end{pmatrix} \boldsymbol{x} = \begin{pmatrix} 0 \\ 0 \\ 5 \end{pmatrix} \Leftrightarrow \boldsymbol{x} = \begin{pmatrix} -1 + \frac{4}{3}t \\ -1 + \frac{5}{3}t \\ t \\ 1 \end{pmatrix}, t \in \mathbb{R}.$$

Question 2(a): Solution

$$\boldsymbol{A} = \begin{pmatrix} 2 & -4 & 4 & -2 \\ 6 & -9 & 7 & -3 \\ -1 & -4 & 8 & 0 \end{pmatrix} \xrightarrow[R_3+\frac{1}{2}R_1]{R_2-3R_1} \begin{pmatrix} 2 & -4 & 4 & -2 \\ 0 & 3 & -5 & 3 \\ 0 & -6 & 10 & -1 \end{pmatrix} \xrightarrow{R_3+2R_2} \begin{pmatrix} 2 & -4 & 4 & -2 \\ 0 & 3 & -5 & 3 \\ 0 & 0 & 0 & 5 \end{pmatrix} = \boldsymbol{R}.$$

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Question 2(b): Solution

$$\boldsymbol{A} = \begin{pmatrix} 2 & -6 & 6 \\ -4 & 5 & -7 \\ 3 & 5 & -1 \\ -6 & 4 & -8 \\ 8 & -3 & 9 \end{pmatrix} \xrightarrow[\substack{R_4+3R_1 \\ R_5-4R_1}]{\substack{R_2+2R_1 \\ R_3-\frac{3}{2}R_1}} \begin{pmatrix} 2 & -6 & 6 \\ 0 & -7 & 5 \\ 0 & 14 & -10 \\ 0 & -14 & 10 \\ 0 & 21 & -15 \end{pmatrix} \xrightarrow[R_5+3R_2]{\substack{R_3+2R_2 \\ R_4-2R_2}} \begin{pmatrix} 2 & -6 & 6 \\ 0 & -7 & 5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \boldsymbol{R}.$$

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Let $\boldsymbol{y} = \boldsymbol{R}\boldsymbol{x}$. Then

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \quad \Leftrightarrow \quad \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 & 0 \\ \frac{3}{2} & -2 & 1 & 0 & 0 \\ -3 & 2 & 0 & 1 & 0 \\ 4 & -3 & 0 & 0 & 1 \end{pmatrix} \boldsymbol{y} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \quad \Leftrightarrow \quad \boldsymbol{y} = \begin{pmatrix} 1 \\ 3 \\ \frac{11}{2} \\ -2 \\ 6 \end{pmatrix};$$

$$\boldsymbol{R}\boldsymbol{x} = \boldsymbol{y} \quad \Leftrightarrow \quad \begin{pmatrix} 2 & -6 & 6 \\ 0 & -7 & 5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \boldsymbol{x} = \begin{pmatrix} 1 \\ 3 \\ \frac{11}{2} \\ -2 \\ 6 \end{pmatrix} \quad \Rightarrow \quad \text{no solution.}$$

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$$\boldsymbol{A} = \begin{pmatrix} 2 & -6 & 6 \\ -4 & 5 & -7 \\ 3 & 5 & -1 \\ -6 & 4 & -8 \\ 8 & -3 & 9 \end{pmatrix} \xrightarrow[\substack{R_4+3R_1 \\ R_5-4R_1}]{\substack{R_2+2R_1 \\ R_3-\frac{3}{2}R_1}} \begin{pmatrix} 2 & -6 & 6 \\ 0 & -7 & 5 \\ 0 & 14 & -10 \\ 0 & -14 & 10 \\ 0 & 21 & -15 \end{pmatrix} \xrightarrow[R_5+3R_2]{\substack{R_3+2R_2 \\ R_4-2R_2}} \begin{pmatrix} 2 & -6 & 6 \\ 0 & -7 & 5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \boldsymbol{R}.$$

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Determinants

The **determinant** of a square matrix $\boldsymbol{A}$,denoted $\det(\boldsymbol{A})$, is a real number assigned to each matrix, with the following characterizing properties:

- If A is upper or lower triangular, then $\det(A)$ is the product of its diagonal entries.
- If B is obtained from A by an application of an elementary row operation, then

$$\det(\boldsymbol{B}) = \begin{cases} -\det(\boldsymbol{A}), & \text{if } \boldsymbol{A} \xrightarrow{R_i \leftrightarrow R_j} \boldsymbol{B}; \\ \det(\boldsymbol{A}), & \text{if } \boldsymbol{A} \xrightarrow{R_i + cR_j} \boldsymbol{B}; \\ c\det(\boldsymbol{A}), & \text{if } \boldsymbol{A} \xrightarrow{cR_i} \boldsymbol{B}. \end{cases}$$

Note

$\det(\boldsymbol{B}) = c\det(A)$ for $\boldsymbol{A} \xrightarrow{cR_i} \boldsymbol{B}$ holds even when $c = 0$.

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- If $\boldsymbol{B}$ is obtained from $\boldsymbol{A}$ by an application of an elementary row operation, then

$$\det(\boldsymbol{B}) = \begin{cases} -\det(\boldsymbol{A}), & \text{if } \boldsymbol{A} \xrightarrow{R_i \leftrightarrow R_j} \boldsymbol{B}; \\ \det(\boldsymbol{A}), & \text{if } \boldsymbol{A} \xrightarrow{R_i + cR_j} \boldsymbol{B}; \\ c\det(\boldsymbol{A}), & \text{if } \boldsymbol{A} \xrightarrow{cR_i} \boldsymbol{B}. \end{cases}$$

Note

$\det(\boldsymbol{B}) = c\det(A)$ for $\boldsymbol{A} \xrightarrow{cR_i} \boldsymbol{B}$ holds even when $c = 0$.

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Computing Determinants Using Gaussian Elimination

Theorem

Suppose that we need k row-pair swaps to obtain a **row echelon form** $\boldsymbol{R}$ *of a square matrix* $\boldsymbol{A}$ *using* **Gaussian elimination**. *Then*

$$\det(\boldsymbol{A}) = (-1)^k \det(R) = (-1)^k \times \textit{ product of diagonal entries of } \boldsymbol{R}.$$

Corollary

Let $\boldsymbol{A}$ be a square matrix. Then $\boldsymbol{A}$ is invertible if and only if $\det(\boldsymbol{A}) \neq 0$.

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Question 3

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Use **Gaussian elimination** to find $\det \begin{pmatrix} 2 & 0 & 3 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 1 & 2 & 3 \end{pmatrix}$.

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Solution.

$$\boldsymbol{A} := \begin{pmatrix} 2 & 0 & 3 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 1 & 2 & 3 \end{pmatrix} \xrightarrow{R_2 - \frac{1}{2}R_1} \begin{pmatrix} 2 & 0 & 3 & 1 \\ 0 & 0 & -\frac{1}{2} & \frac{1}{2} \\ 0 & 2 & 1 & 0 \\ 0 & 1 & 2 & 3 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 2 & 0 & 3 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & -\frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 2 & 3 \end{pmatrix}$$

$$\xrightarrow{R_4 - \frac{1}{2}R_2} \begin{pmatrix} 2 & 0 & 3 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & \frac{3}{2} & 3 \end{pmatrix} \xrightarrow{R_4 + 3R_3} \begin{pmatrix} 2 & 0 & 3 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & \frac{9}{2} \end{pmatrix} =: \boldsymbol{R}.$$

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Thus $\det(\boldsymbol{R}) = -\det(\boldsymbol{A})$, so that

$$\det(\boldsymbol{A}) = -\det(\boldsymbol{R}) = -(2)(2)(-\tfrac{1}{2})(\tfrac{9}{2}) = 9.$$

□

Next Lesson ...

We will explore **Sections 2.8–2.10**.
Please watch the video clips for these sections before then.